

SAM BAI

PRODUCT ENGINEER | FULL-STACK SYSTEMS

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PROFILE

Product engineer and founder who turns ambiguous workflows into production software across discovery, interaction design, architecture and operations. Shipped a role-aware construction platform for a New Zealand building team and built Trekky, a job-search operating system spanning web, PWA, mobile, browser extension, AI and MCP surfaces.

EXPERIENCE

Founder & Product Engineer - Solynth Labs Limited

2026 - Present

Auckland, New Zealand | Software consultation, production and owned products

- Lead end-to-end product delivery across client and owned products: clarify workflows, prototype risky interactions, define data and security boundaries, ship releases and operate the result.
- Shipped Rivet, a production construction programme and exception-management platform for a New Zealand residential building team, with role-specific mobile and desktop workflows, schedule approvals, local-first outbox sync, push/calendar integrations, and private evidence and voice capture.
- Built Rivet on React 19/Vite, Capacitor, Hono/Cloudflare Workers, Neon PostgreSQL via Hyperdrive and R2; the protected release passed 444 web/API tests plus real-Postgres, two-seat sync, accessibility, migration and production smoke gates.
- Built and operate Trekky v1.11, a private job-search OS covering role discovery, lifecycle tracking, AI apply kits, outcome analytics, Google sync, PWA/share capture, a browser extension and account-scoped MCP; Next.js 16, TypeScript, MongoDB, Redis, Better Auth and Gemini.

Software Engineering Intern - Waka Transportation Community

Mar - Jul 2024

Tauranga, New Zealand | Community transport platform

- Delivered MERN features from requirements through testing and stakeholder demonstration, including user authentication and a driver-scheduling calendar for a community transport service.
- Built responsive React components and validated REST endpoints, improved component performance, and translated weekly feedback from non-technical stakeholders into scoped releases.

Career Break & Independent Product Development - Self-directed

Aug 2024 - Dec 2025

New Zealand and overseas

- Balanced family commitments and travel with deliberate full-stack practice; built TaskFlow, a multi-tenant collaboration SaaS using Supabase RLS/Realtime, Clerk, Stripe and optimistic drag-and-drop.

SELECTED ENGINEERING WORK

Clearfold (2026) - Architected a pre-production digital-asset operations system with immutable PostgreSQL evidence, role-scoped controls, reconciliation and risk workflows, and Vitest/Playwright/PGLite verification.

RentAKL (2026) - Designed a map-first Auckland rental-intelligence prototype using Next.js, PostGIS/RLS, MapLibre, isolated ingestion workers and deterministic listing-risk scoring.

EDUCATION

Bachelor of Applied Information Technology (Software Engineering) - Waikato Institute of Technology (Wintec)2024

Hamilton, New Zealand

- Led a four-person Kotlin/Android medication-tracking capstone for Anglesea Clinic; owned planning, code reviews and the clinical stakeholder demonstration.

SKILLS

Product: Discovery, workflow mapping, interaction design, accessibility, systems design, production operations

Frontend & mobile: TypeScript, React, Next.js, Vite, Tailwind CSS, PWA, Capacitor, Flutter

Backend & data: Node.js, Hono, REST/tRPC, PostgreSQL (Neon/Supabase), MongoDB, Redis, RLS, local-first sync

Cloud, quality & AI: Cloudflare Workers/R2/Hyperdrive, Vercel, GitHub Actions, Vitest, Playwright, Docker, Gemini/OpenAI, MCP